Sparse Quantum Simulation

CSC591/ECE592 Project Update #1

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Prompt: Report on the progress of your project and update your web page. This includes a report on solved issues, indicating individual contributions by each group member by his/her name and a corresponding milestone/task, as well open problems and the steps for attacking them for the remainder of the project (indicating weekly milestones of progress). Turn in the file report2.pdf. Make sure to update your web page.

I. PROGRESS

Much of the beginning week (or two) of the project was spent ensuring that we were on the same page of project expectations (particularly choice of programming language) and the end (and future) goal of the project. From there, discussions began on how to go about actually representing gates under the assumption that CSC, CSR, and COO representations are off limits. Our initial idea was to create a collection of diagonals, each of which would contain start/end index pairs - using typical i, j indexing of matrices - and the data stored in all points between those index pairings. While this approach would likely work, we discovered that it would make matrix multiplication difficult. So we pivoted and started looking into options which could sparsely represent an entire principle/anti- diagonal.

Our original thought was that there would be some cases where it could be ideal to have a combination of the two: consider a toffolli gate, it could be represented by the combination of principle diagonal and one anti-diagonal. However, given the time constraints of the project we elected to stick with main diagonal. By doing this, the bulk of the work is handled by a shift operation and by an element-wise multiplication.

Our largest core assumption is that all quantum gates can be represented as a matrix and therefore the 'tensor' contractions we'll be working with simplify to only needing the Kronecker product and product of matrices. These matrices are also assumed to be sparse and contain entries mostly on diagonals.

Under these assumptions, we have found that in the worst case our matrix-matrix algorithm has a amortized time complexity of O(n*d*d) where n is the length of diagonal (¡=n), d is the number of diagonals (equal to 2n-1 for a square matrix of size n).

Also, please take a look at our website for some preliminary results and analysis.

II. INDIVIDUAL CONTRIBUTIONS

Item (in chronological order)	Blake	Srikar
Research Existing Sparse Representations	x	x
Discussions on Sparse Structure Exploitation	x	x
Discussion with Dr. Mueller (adjusting focus)		x
Work through initial thoughts on 'sparse diags' representation	x	x
Build shell of Vector class (Dense)	x	
Build shell of sparse Matrix class		x
Discuss Mat-Mat product algorithm and complexity	x	x
Github/Web page point person		x
LaTeXpoint person	x	

Table I: Summary of contributions made thus far

III. OPEN PROBLEMS BEING TACKLED

Our biggest issue right now is leveraging our sparse representation to produce a mat-mat multiplication algorithm which can be accurately applied to our sparse diagonals representation in a manner which actually exploits sparsity.

IV. UPDATED TIMELINE

Week of...	Work
Oct 31	- Working out many algorithmic hurdles.
	- Update approach to only use Principle diagonals (for now)
Nov 7	- Design new algorithm, analyze complexities and scope for parallelism.
	- Documentation, progress reports.
Nov 14	- Once algorithms checked, build out the kernels in C++.
Nov 21	- Compare #flops, memory (kB) and #intops of custom solution to existing solutions.

Table II: Ever-changing Schedule